

## Part A · Multiple Choice

2 points each

1. **a) the galaxy is moving away from Earth**

*p. 24 — redshift occurs when the light source moves away from the observer, or when the space between them stretches. Because all galaxies show redshift, the universe must be expanding.*

2. **d) the farther away a galaxy is, the faster it is moving away from us**

*p. 24 — Hubble combined distance measurements with redshift data and found this relationship, which leads to the hypothesis that the universe is expanding.*

3. **b) 13.7 billion years ago**

*p. 25 — 4.6 billion years is the age of the solar system, not the universe. 380,000 years is how long after the Big Bang the first neutral atoms formed.*

4. **c) nuclear fusion**

*p. 33 — in the core of a star, pressure is great enough that two hydrogen atoms fuse to become a helium atom.*

5. **a) globular cluster**

*p. 29 — open clusters, by contrast, hold up to a few thousand young bluish stars loosely, and still contain glowing gas and dust.*

6. **b) 26,000 light-years**

*p. 32 — a little more than halfway out from the center to the edge. The disk itself is about 100,000 light-years across.*

## Part B · True or False

1 point each

7. **TRUE**

*p. 30 — most elliptical galaxies also contain very little gas and dust, because it has already formed into stars.*

8. **FALSE**

*The Milky Way is a typical spiral galaxy, with a disk, a central bulge, and spiral arms. (p. 31)*

## Part C · Short Answer

3 points each

9. **Dark matter accounts for missing mass; dark energy accounts for the accelerating expansion of the universe.**

**Rubric:** 1.5 pts each. **Dark matter:** there is far too little visible matter to make up the roughly 27% of the universe that observations require, so dark matter is the name for that unseen mass. Scientists are more certain what it is *not* — not stars and planets, not baryonic clouds, not antimatter, not galaxy-sized black holes. **Dark energy:** a hypothetical form of energy, possibly as much as 72% of the universe's total energy content, proposed to explain why the rate of expansion is *increasing* rather than slowing. (pp. 26–28)

10. **Fusion stops completely, and gravitational collapse shrinks the star's core into a white dwarf.**

**Rubric:** 1 pt fusion stops; 1 pt gravitational collapse of the core; 1 pt the result is a white dwarf — a white, glowing object about the size of Earth — which eventually fades out. Award credit if the student also describes the earlier red giant stage (outer layers grow outward and cool, turning the star red). Do *not* give credit for supernova, neutron star, or black hole — those require a much more massive star. (p. 36)